HACK MATRIX HACKATHON

- **PROJECT TITLE**

  : Carpool Finder

- **TEAM NAME**

  : Neon Syntax

- **TEAM MEMBERS :**

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PROBLEM STATEMENT

- **Problem Description:-** Urban commuting is dominated by single-occupancy vehicles, leading to massive traffic congestion and excessive fuel waste.

- **Who is affected by this problem:-** Daily office commuters (e.g., Bhopal to Indore city).
  - University students traveling long distances.
  - Urban residents facing rising fuel prices.

- **Why this problem is important ?**

- Transportation accounts for a significant portion of CO2emissions. Reducing the number of cars on the road is critical for both financial savings and environmental health.

EXISTING SOLUTIONS AND GAPS

- **List of existing Solutions**
  - Public Transport: Often inflexible and overcrowded.
  - Ride-Hailing Apps (Uber/Ola): Extremely expensive for daily commutes, even it is not a solution its just a taxi option which isn't helping in lowering Carbon Emissions.

- **Mention their Limitations -**
  - No specialized system for long-term route matching.
  - Lack of social trust/safety features for carpooling with strangers.
  - No feedback loop on environmental impact.

- **Explain the gap your project will address**
  - WayShare bridges the gap by providing a **low-cost**,sustanability platform that motivate by showing amount of co2 it save .

PROPOSED SOLUTION

- **Explain your solution clearly-**
  - Web-based carpooling dashboard where users seamlessly offer and book seats.

- **List key features**
  - **Smart Fare-Split:** Automatically calculates the share per person (Total Cost ÷ Total Seats).
  - **Emission Tracker:** Calculates Co2 saved per ride using an average factor of 0.46$ kg/km.

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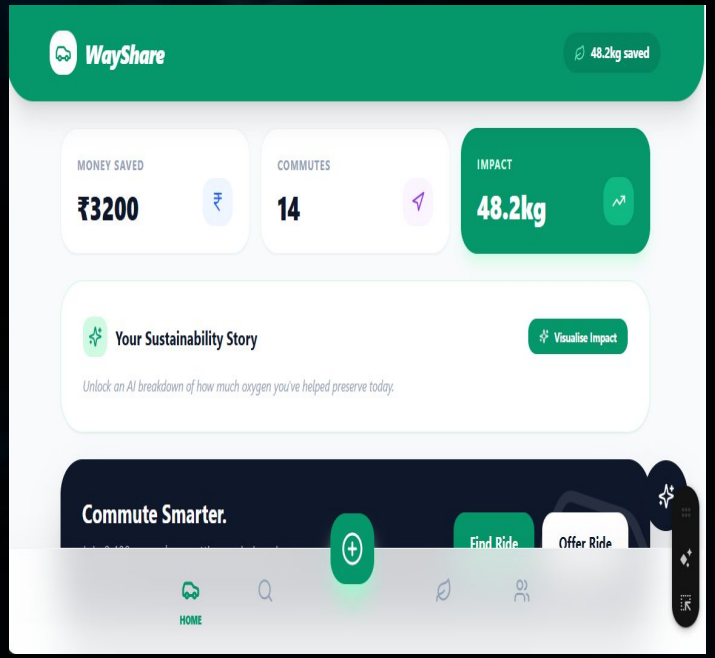
SYSTEM ARCHITECTURE

- **Frontend:** React.js for a dynamic, single-page application experience.
- **Logic Layer:** State management handles ride listings, booking math, and local stats.
- **Intelligence Layer:** Google Gemini API processes user data for smart content.
- **Data Flow:** 1. User Input (Route/Time) rightarrow 2.Logic Layer (Calculation/Matching) 3. Gemini API (Bio/Icebreaker Enrichment) 4. UI Rendering

Note:(for javascript part we take help of google gemini)



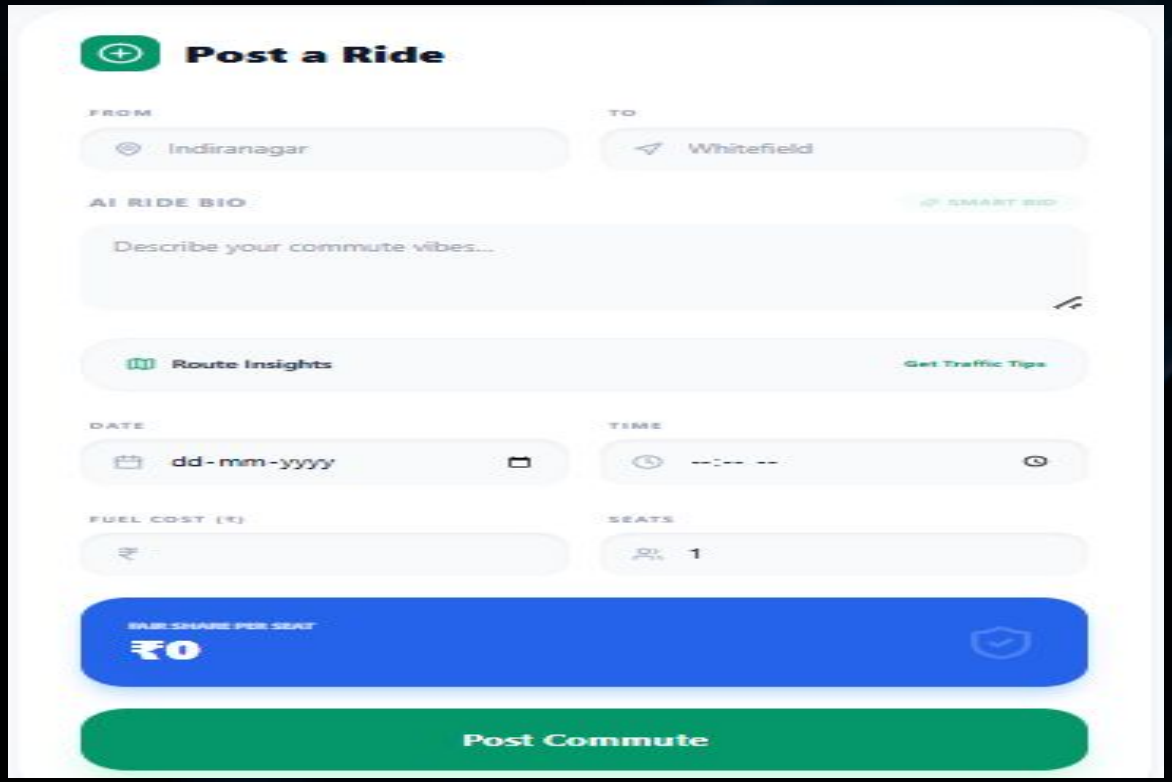
FLOWCHART / WORKING PROCESS

- **Show step-by-step working -**
  - **Registration:** User creates a verified profile.
  - **Offering:** Driver posts a ride;
  - **Discovery:** Rider searches for destination; UI filters available matches.
  - **Booking:** Rider joins; Smart-Split calculates fare; Stats update.
  - **Engagement:** Users get "Journey Icebreakers" to foster a social environment.

TECH STACK

- **Frontend:** React.js — Used for modular component architecture and efficient rendering.
- **Styling:** Tailwind CSS — Provides a mobile-first, responsive, and modern "Emerald" UI.
- **Icons:** Lucide-React — Comprehensive icon library for intuitive navigation.
- **State Management:** React Hooks (useState)— Manages real-time data without external heavy libraries.

IMPLEMENTATION PLAN



EXPECTED OUTCOME / IMPACT

Expected Results: A fully functional platform that turns a solo drive into a social, cost-saving event.

User Benefits:

- Up to 75% reduction in daily commute costs.
  - Reduced stress through social interaction and AI-guided routes.

Real-world Impact:

Measurable reduction in regional CO2 footprints (e.g., 42.5kg saved over 12 rides shared).

FUTURE SCOPE AND CONCLUSION

- **Future Scope:** * Real-time GPS tracking and route optimization.
  - Secure In-App payment gateway integration.
  - award: Rewards/Badges for "Top Green Commuters."
- **Conclusion:**

  WayShare isn't just a carpool app; it’s a community-driven move toward a sustainable future.